

① fie $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $f(x) = (-x_1 - 2x_3, x_1 + x_2 + x_3, x_1 + 2x_3)$

o) determinați $\ker f$ și $\operatorname{Im} f$

• La exercițiul de genul este bine să aduceți expresia lui f la o formă $f(x) = A \cdot x$

$$\begin{aligned} f(x) &= (-x_1 - 2x_3, x_1 + x_2 + x_3, x_1 + 2x_3) \\ &= (-x_1, x_1, x_1) + (0, x_2, 0) + (-2x_3, x_3, 2x_3) \\ &= x_1(-1, 1, 1) + x_2(0, 1, 0) + x_3(-2, 1, 2) \end{aligned}$$

$$\Rightarrow f(x) = \begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\ker(f) = \left\{ x \in \mathbb{R}^3 \mid f(x) = 0_{\mathbb{R}^3} \right\}$$

$$f(x) = 0_{\mathbb{R}^3} \Rightarrow Ax = 0_{\mathbb{R}^3}$$

- Do cō $\text{rg } A = 3 \Rightarrow \text{SCD} \Rightarrow X = 0 \Rightarrow$
 $\Rightarrow \ker(\varphi) = \{0_{\mathbb{R}^3}\}$

- Do cō $\text{rg } A < 3 \Rightarrow \text{SCN} \Rightarrow$
 $\Rightarrow \dim \ker(\varphi) = 3 - \text{rg } A$

$$A = \begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}$$

$$\det(A) = 1 \cdot (-1)^{2+2} \cdot \begin{vmatrix} -1 & -2 \\ 1 & 2 \end{vmatrix} = 0 \Rightarrow$$

$$\Rightarrow \text{rg}(A) < 3$$

$$\Delta_P = \begin{vmatrix} -1 & 0 \\ 1 & 1 \end{vmatrix} = -1 \neq 0 \Rightarrow x_1, x_2 - \text{m.e.c.} \\ \text{principale}$$

$$AX = 0$$

$$\Rightarrow \begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$-x_1 - 2x_3 = 0 \Rightarrow x_1 = -2x_3$$

$$x_1 + x_2 + x_3 = 0 \Rightarrow x_2 = -3x_3$$

$$\ker(f) = \{x \in \mathbb{R}^3 \mid Ax = 0\} =$$

$$\Rightarrow \ker(f) = \{(-2x_3, -3x_3, x_3) \mid x_3 \in \mathbb{R}\}$$

$$\ker(f) = \langle (-2, -3, 1) \rangle$$

$$\operatorname{Im} f = \{y \in \mathbb{R}^3 \mid \exists x \in \mathbb{R}^3 \text{ mit } f(x) = y\}$$

$$f(x) = y \Rightarrow Ax = y$$

• Also wenn es ein System so

für uns kompatibel

$$\operatorname{rang}(A) = 2 \Rightarrow \operatorname{rang}(\bar{A}) = 2 \Rightarrow$$

$$\Rightarrow \det(\bar{A}) = 0$$

• Also wenn $y = (y_1, y_2, y_3)$ mit $\det(\bar{A}) = 0$

$$\begin{vmatrix} -1 & 0 & y_1 \\ 1 & 1 & y_2 \\ 1 & 0 & y_3 \end{vmatrix} = 0 \Rightarrow 1 \cdot (-1)^4 \cdot \begin{vmatrix} -1 & y_1 \\ 1 & y_3 \end{vmatrix} = 0$$

$\Rightarrow$

$$\Rightarrow \begin{vmatrix} -1 & y_1 \\ 1 & y_3 \end{vmatrix} = 0$$

$$-y_3 - y_1 = 0 \Rightarrow y_1 = -y_3$$

$$\Rightarrow \text{Im}(f) = \{ (-y_3, y_2, y_3) \mid y_2 \in \mathbb{R}, y_3 \in \mathbb{R} \}$$

$$\Rightarrow \text{Im}(f) = \langle (-1, 0, 1), (0, 1, 0) \rangle$$

b) stim c\o \$f(x) = Ax\$

in baza canonic\o, coordonatele lui \$x\$ sunt chiee \$(x_1, x_2, x_3) \Rightarrow\$

\$\Rightarrow\$ si coordonatele lui \$v = Ax\$ vor fi

$$\text{not } (v_1, v_2, v_3) = v \Rightarrow [f]_{R_0 R_0} = A$$

Care este matricea aplicatiei liniare
 in raport cu baza $B = \{(1, 3, -1), (1, 4, 0),$
 $(1, 2, -1)\}$
 $(1, 0, 0) =$

$$R_0 \xrightarrow{C} B$$

$$X = C X' \quad , \quad f(X) = C f(X')$$

$$f(X) = A C X'$$

$$C f(X') = A C X'$$

$$f(X') = C^{-1} A C$$